

Blake Porteous

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Education

University of Auckland – BE (Honours), Engineering

Expected 2029

- Young Scholars MAX199
- Completed 3 university-level mathematics papers while in high school
- Studied topics including cryptography, graph theory, linear algebra, data analysis, and multivariable calculus
- Currently taking second-year mathematics papers

University of Canterbury – STAR MATHS199 (A+)

December 2025

- 2 first-year engineering maths papers at University of Canterbury

Skills

Languages: Python, Swift, C++

Software: Matlab, Firebase, XCode, PyTorch, Fusion, RStudio

Projects

Tasked – Social Goal Tracking App

- iOS app that assigns users shared tasks and tracks completion
- Developed a full-stack app using SwiftUI with Firebase Authentication and Firestore
- Implemented a system where users receive tasks and submit completion progress
- Designed real-time data syncing to track user activity and consistency
- Built and iterated on the app based on personal use and testing

Retro Space Invaders

- Recreated a classic Space Invaders-style game in C++ targeting Windows XP
- Built core game systems including rendering, input handling, and collision detection
- Worked with C++ project structure, libraries, and directory management in Visual Studio
- Focused on replicating retro gameplay mechanics within legacy system limitations

Neural Network

- Learnt how NNs work by building one from scratch in NumPy with no ML libraries
- Wrote the dense layers, ReLU and Softmax activations, and cross-entropy loss myself
- Moved to PyTorch and trained a convolutional network to recognise handwritten digits (MNIST)
- Reached 97% accuracy on the 10,000-image test set

Strengthening Democracy – Youth Engagement Campaign

- Developed a youth-focused campaign to increase political engagement
- Produced a targeted video campaign promoting youth participation in politics
- Incorporated insights from student leaders and policymakers to refine messaging
- Presented the campaign within a wider initiative involving senior leaders and public figures

Experience

Football Coach, Onslow Junior Football Club – Wellington Mar 2022 – Nov 2025

- Led structured football training sessions for children aged 5–10
- Organised drills, equipment, and session flow to ensure smooth and efficient practices
- Maintained high levels of organisation and preparedness to manage group safety and engagement

Math Tutor, Self-employed – Wellington Feb 2025 – Nov 2025

- Provided one-on-one maths tutoring, explaining concepts clearly and step-by-step
- Adapted teaching style to suit different ability levels and learning needs
- Built patience and communication skills through regular student support

Volunteer Math Tutor – Wellington Mar 2025 – Nov 2025

- Volunteered in Saturday tutoring sessions helping NCEA Level 2, Level 3, and Scholarship Calculus
- Assisted with challenging problem-solving questions and exam preparation

Competitions

Maths: Computational Algorithmic Thinking – High Distinction (Aced); Australian Maths Competition – Distinction; Mathswell – 1st (2020, 2021), 2nd (2022)

Physics: 1st Place, Junior Youth Physics Tournament

Interests

Football, Tennis, Rock Climbing, Chess, Skiing